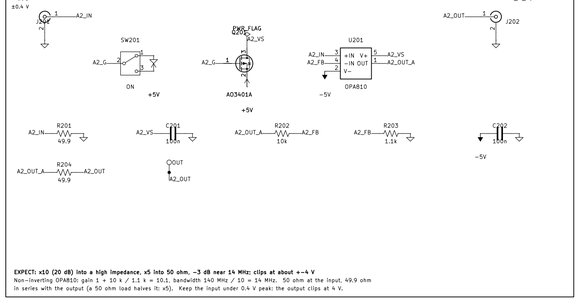
## A2 x10, 14 MHz

tutorial 4.06, 5.02 (+−5 V)

cell (0,0)

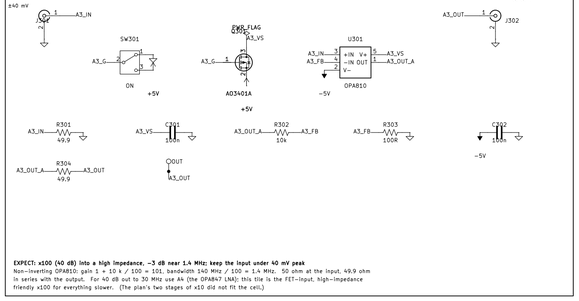



**EXPECT: x10 (20 dB) into a high impedance, x5 into 50 ohm, −3 dB near 14 MHz; clips at about +−4 V**
Non-inverting OPA810: gain 1 + 10 k / 1.1 k = 10.1, bandwidth 140 MHz / 10 = 14 MHz. 50 ohm at the input, 49.9 ohm in series with the output (a 50 ohm load halves it, x5). Keep the input under 0.4 V peak: the output clips at 4 V.

## A3 x100, 1.4MHz

tutorial 4.06, 5.02 (+−5 V)

cell (1,0)

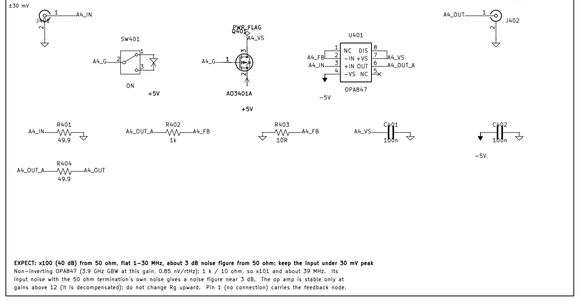



**EXPECT: x100 (40 dB) into a high impedance, −3 dB near 1.4 MHz; keep the input under 40 mV peak**
Non-inverting OPA810: gain 1 + 10 k / 100 = 101, bandwidth 140 MHz / 100 = 1.4 MHz. 50 ohm at the input, 49.9 ohm in series with the output. For 40 dB out to 30 MHz use A4 (the OPA847 LNA); this tile is the FET-input, high-impedance friendly x100 for everything slower. (The plan's two stages of x10 did not fit the cell.)

## A4 LNA, 30 MHz

tutorial 4.06, 5.03 (+−5 V)

cell (2,0)

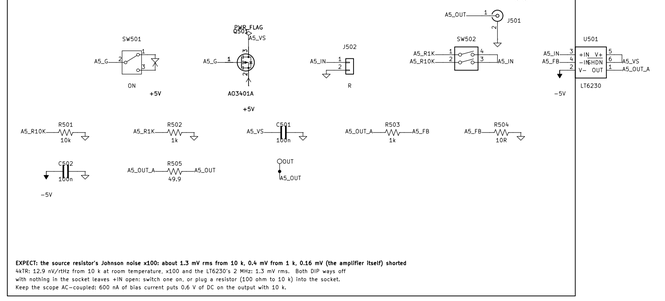



**EXPECT: x100 (40 dB) from 50 ohm, flat 1–30 MHz, about 3 dB noise figure from 50 ohm; keep the input under 30 mV peak**
Non-inverting OPA847 (3.9 GHz GBW at this gain, 0.85 nV/√Hz); 1 k / 10 ohm, so x101 and about 39 MHz. Its input noise with the 50 ohm termination's own noise gives a noise figure near 3 dB. The op amp is stable only at gains above 12 (it is decompensated); do not change Rg upward. Pin 1 (no connection) carries the feedback node.

## A5 noise, x100

tutorial 5.08, 7.05 (+−5 V)

cell (3,0)

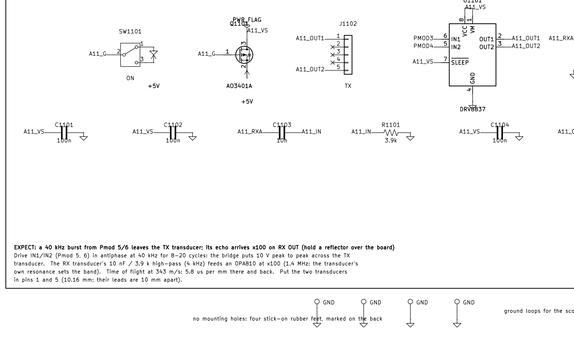



**EXPECT: the source resistor's Johnson noise x100; about 1.3 nV rms from 10 k, 0.4 nV from 1 k, 0.16 nV (the amplifier itself) shorted**
4kTR: 12.9 nV/√Hz from 10 k at room temperature, x100 and the LT6230's 2 MHz: 1.3 nV rms. Both DIP ways off with nothing in the socket leaves +IN open: switch one on, or plug a resistor (100 ohm to 10 k) into the socket. Keep the scope AC-coupled: 600 nA of bias current puts 0.6 V of DC on the output with 10 k.

## A6 1-bit ADC

tutorial 7.05 (+3.3 V)

cell (0,1)

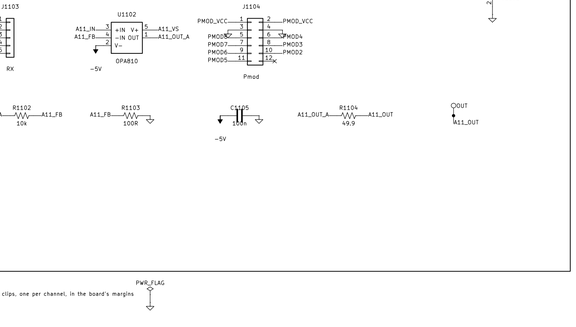



**EXPECT: a one-bit stream on Pmod 2 whose density follows IN: the FPGA closes the loop through Pmod 1 (the handbook's gateware)**
First-order delta-sigma: the comparator decides, the FPGA registers the bit and feeds it back through 10 k into the 10 k / 10 nF integrator (1.6 kHz; clock the loop at 1–10 MHz). IN takes 0 to 3.3 V, or +−1.5 V about the 1.65 V reference if you AC-couple. Both Pmod lines are 3.3 V logic; the tile runs on the 3.3 V rail.

## A7 PLL 4046

tutorial 4.01 (+5 V)

cell (1,1)

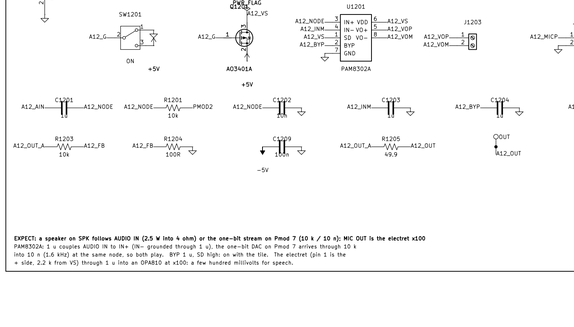



**EXPECT: VCO OUT locks to a 50–200 kHz square or sine on IN; pin 9 is the demodulated FM, pin 13 the phase detector**
CD74HC4046A: R1 10 k, C1 1 nF put the VCO near 100 kHz (R2 open; it can go down to 0). PC2 (the edge-triggered tri-state detector) through 10 k / 100 n into VCOIn: a lag filter, so expect ringing on lock. IN self-biases through 100 n / 1 k (any level above 0.4 V peak). VCO OUT is 5 V CMOS through 470 ohm. Probe pins 9 and 13.

## A8 heat

tutorial 5.02, 7.06 (+5 V)

cell (2,1)

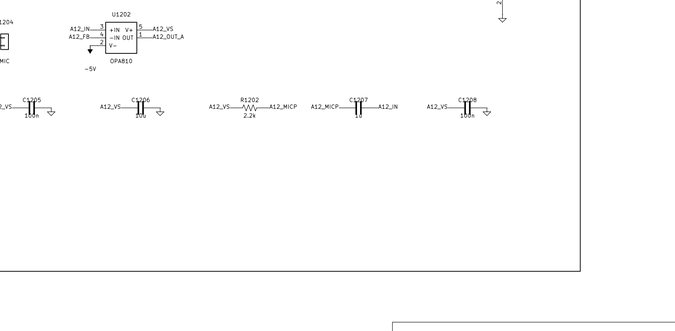



**EXPECT: 0.83 W into the board when Pmod 3 is high (the FET's drain), 10 mV/K + 500 mV on OUT**
Two 15 ohm in series: 30 ohm, 170 mA, 0.83 W from the switched 5 V when Pmod 3 is high (10 k keeps it off otherwise). The TMP235 reads 750 mV at 25 C and climbs 10 mV per kelvin. A feedback-control experiment: hold the TMP235 at a set point with the FPGA's PWM. (The plan's NTC divider did not fit the cell beside the two 2512s.)

## A1 power, USB-C 5 V in: +5, −5, +3.3 out

tutorial 1.01

cell (3,1)




**EXPECT: +5 V from the USB-C (up to 1.5 A), −5 V at 40 mA, +3.3 V at 300 mA; the green LED says 5 V is in, the red one that −5 V is up**
USB-C power only: 5.1 k on each CC asks any source for 5 V. A 2 A PTC, a 5 V TVS (clamps at 9.2 V) and 20 uF. The LM2664 charge pump makes −5 V for the op-amp tiles (40 mA; four tiles at 10 mA): it is not switched per tile, an op amp with its +5 V removed draws nothing. The TPS7A2033 makes 3.3 V for the comparator and the lines that meet the FPGA. The socket along the top carries +5, ground, −5, ground, +3.3; the bench's rails for a breadboard experiment.

## A9 mixer, MC1496

tutorial 1.10, 6.02 (+−5 V)

cells (0,2), (1,2)




**EXPECT: IF OUT = RF x LO (sum and difference frequencies, about 0.3 V peak per volt of RF at 0.3 V of LO); a balanced mixer, the carrier mostly cancels**
Gilbert cell on +−5 V: signal pair at ground (51 ohm), carrier pair at 2.5 V (51 ohm to the 10 k / 10 k bias), 1 mA per side (3.9 k), 1.5 k loads. Keep RF under 0.3 V and LO near 0.3 V peak (above that the carrier pair just switches, which is fine for a product detector). IF OUT is single-ended from pin 6 through 100 n: a 1.5 k source, so read it with a high-impedance input, not 50 ohm. No carrier-null trimmer: expect −40 dB of LO feedthrough.

## A10 light, LED + PD

tutorial 4.07 (+−5 V)

cells (2,2), (3,2)




**EXPECT: 45 mA of LED current per volt on DRIVE IN (0–1 V, blanked while Pmod 4 is high); PD OUT climbs 10 V per mA of photocurrent**
The current source: the op amp holds 22 ohm at the input voltage, so the LED carries V / 22 ohm (0.9 V is the LED's full 40 mA; above 1.1 V the op amp runs out of headroom). A 2N7002 from Pmod 4 shorts the input: TTL modulation to a few MHz. The photodiode (5 V reverse) into 10 k with 10 pF: 1.6 MHz; daylight on the PD gives millivolts, the LED of a second board across a desk gives tens. The LED looks out of the board's right edge, the photodiode looks up.

## A11 ultrasound

tutorial 4.06 (+−5 V)

cells (0,3), (1,3)




**EXPECT: a 40 kHz burst from Pmod 5/6 leaves the TX transducer; its echo arrives x100 on RX OUT (hold a reflector over the board)**
Drive IN1/IN2 (Pmod 5, 6) in antiphase at 40 kHz for 8–20 cycles: the bridge puts 10 V peak to peak across the TX transducer. The RX transducer's 10 nF / 3.9 k high-pass (4 kHz) feeds an OPA810 at x100 (1.4 MHz: the transducer's own resonance sets the band). Time of flight at 343 m/s: 5.8 us per mm there and back. Put the two transducers in pins 1 and 5 (10.16 mm: their leads are 10 mm apart).

## A12 audio

tutorial 1.05, 6.04 (+−5 V)

cells (2,3), (3,3)




**EXPECT: a speaker on SPK follows AUDIO IN (2.5 V into 4 ohm) or the one-bit stream on Pmod 7 (10 k / 10 n): MIC OUT is the electret x100**
PAM8302A: 1 u couples AUDIO IN to IN+ (IN− grounded through 1 u), the one-bit DAC on Pmod 7 arrives through 10 k into 10 n (1.6 kHz) at the same node, so both play. BYP 1 u, SD high: on with the tile. The electret (pin 1 is the + side, 2.2 k from VS) through 1 u into an OPA810 at x100: a few hundred millivolts for speech.

GND GND GND GND

no mounting holes: four stick-on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

PWR_FLAG

Drawn by scripts/gen_sch.py from boards/amod_a.py: the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
gain blocks, an HF LNA, Johnson noise, a one-pin ADC, a 74HC4046 PLL, heat, a mixer, light, ultraso
**Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/**

Sheet: /
File: amod_a.kicad_sch

**Title: AMOD-A: active building blocks on USB-C power**

Size: A1 | Date: 2026-10-25 | **Rev: A**
KiCad E.D.A. 10.0.7 | Id: 1/1